

DIPU BADATYA

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Final-year Computer Science student with hands-on experience in full stack development using the MERN stack. Built and deployed scalable applications featuring real-time communication, geospatial search, and secure authentication systems. Focused on writing clean, maintainable code and solving real-world problems through practical implementations. .

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, HTML5, CSS3
- **Frontend:** React.js, Tailwind CSS, EJS, Responsive Web Design
- **Backend:** Node.js, Express.js, REST APIs, JWT, Passport.js, Bcrypt
- **Database:** MongoDB, Mongoose, Aggregation, Geospatial Indexing (2dsphere)
- **Tools & Platforms:** Git, GitHub, Socket.io, Postman, Vercel, Render, VS Code

PROJECTS

LifeLink – Emergency Blood Donor Discovery Platform

[\[GitHub\]](#) | [\[Live\]](#)

React.js, Node.js, Express.js, MongoDB, Socket.io, Leaflet.js

- **Built** a full stack platform that connects hospitals with nearby blood donors in real time during medical emergencies.
- **Implemented** location-based donor search using **MongoDB 2dsphere indexes**, allowing filtering within a configurable 1–20 km radius.
- **Designed** a secure login system with **JWT authentication** and **Role-Based Access Control** to separate Donor and Hospital dashboards.
- **Enabled** real-time emergency alerts using **Socket.io**, allowing hospitals to instantly notify matched donors without page refresh.
- **Visualized** donor locations on an interactive map using **Leaflet.js** with OpenStreetMap for intuitive geographic navigation.

Real-Time College Bus Tracking System

[\[GitHub\]](#) | [\[Live\]](#)

React.js, Node.js, Express.js, MongoDB, Socket.io, Leaflet.js, Tailwind CSS

- **Developed** a live bus tracking web application that broadcasts real-time GPS updates to **500+ student users** via WebSockets.
- **Displayed** dynamic bus movement on an interactive dashboard by integrating **Leaflet.js** with OpenStreetMap tile layers.
- **Organized** the backend codebase using the **MVC architecture**, improving code readability and making routes modular and testable.
- **Deployed** the application on **Render** with environment-variable-based configuration for smooth development-to-production workflow.

Diary – Social Writing and Real-Time Chat Platform

[\[GitHub\]](#) | [\[Live\]](#)

Node.js, Express.js, MongoDB, Socket.io, EJS

- **Created** a social blogging platform with features for publishing posts, commenting, following users, and private messaging.
- **Secured** user accounts using **Bcrypt** for password hashing, server-side sessions, and automated email verification on signup.
- **Designed** efficient MongoDB schemas with proper indexing to handle concurrent reads and writes on user activity feeds.
- **Added** real-time notifications for likes and follows using **Socket.io**, increasing user interaction and platform engagement.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2022 – 2026

Roland Institute of Technology, Brahmapur, Odisha

CGPA: 8.0 / 10.0

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems,

ADDITIONAL

- Achieved **Google Arcade Facilitator Program – Legend Tier**
- Built and maintained **3 major projects** with real-world features and scalability focus